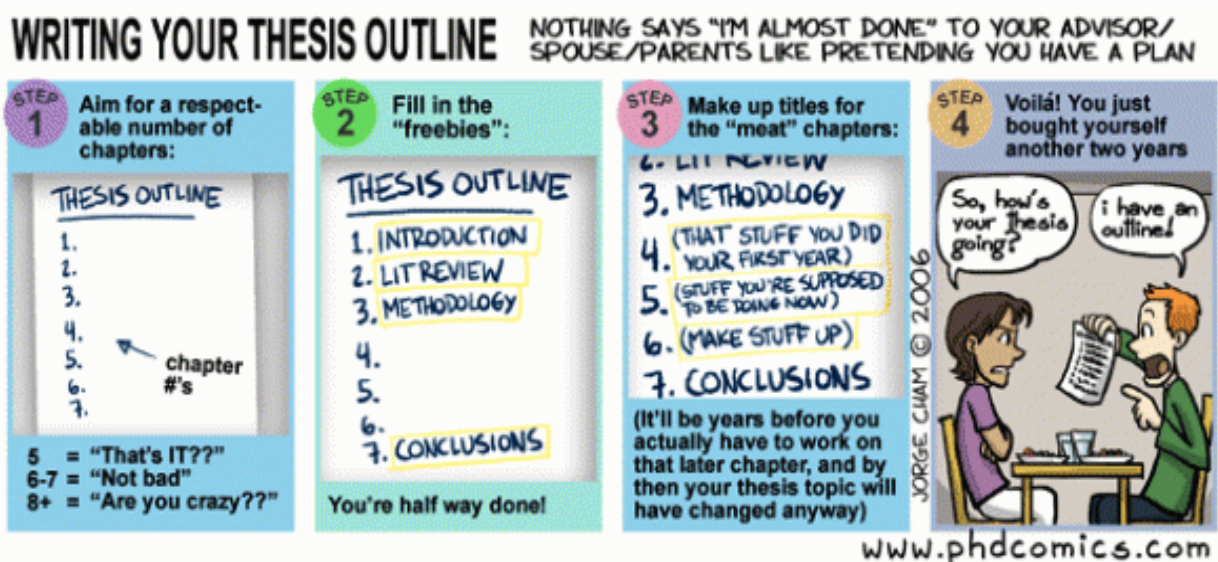


Swiss TPH Scientific writing

Lesson 4: Proposal structure and writing the Introduction section of a research paper/thesis



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Table of Contents

Lesson 4: Proposal structure and writing the Introduction section of a research paper/thesis	1
1. Objectives lesson 4	2
2. Proposal writing	2
2.1 Structure of a research proposal	3
3. The Introduction section of a research article or thesis	5
3.2 Establishing a niche: Gap/Problem statement	7
3.3 Occupying the niche: Filling the gap/addressing the problem	11
4. The hypothesis / research questions	13
5. References	16

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1. Objectives lesson 4

In this lesson, we will focus on:

- Why one should write a proposal. Structure. Tips.
- Writing the Introduction section of a thesis or research article
- Including the three moves in research introductions
- Establishing a territory: centrality statement
- Establishing a niche: writing a gap/problem statement
- Occupying the niche/addressing the problem
- Formulating a Hypothesis and Research Questions

2. Proposal writing

A well-written research proposal enables the university or funding institution to evaluate your suitability for graduate level research, decide if your research is new and original, assess if financial support for your proposed area of study can be found, and identify the right supervisor for you. Your research proposal may vary depending on the subject area your PhD will cover. Broadly speaking, arts and social sciences disciplines will require a different type of proposal to engineering, science and health sciences disciplines.

The actual work you will undertake during the course of your research may differ from your proposal. However, you will need to demonstrate an understanding of the issues associated with conducting research and the potential to design a research study to address a specific set of issues.

You are not expected to have all the answers, but a successful proposal should:

- clearly define your intended topic
- demonstrate that you understand your research area
- indicate that you have started to develop an original and interesting research question
- show that you understand how to conduct research
- look professional – ensure your proposal is well-structured, clear and legible, and that you have checked for any spelling or grammar errors
- focus on quality rather than quantity

The online Swiss TPH Handbook contains a template and clear guidelines on writing a PhD proposal.

2.1 Structure of a research proposal

A research proposal follows the same structure as a research article or thesis, except, of course, you do not yet have results that you can present and discuss, or conclusions that you can draw.

- **Title**– this is the working title. It should be clear, concise and describe the theme of your proposal.
- **Overview/Abstract**– include a short summary of your proposal, stating the problem or question you plan to address, the methods you will follow, and the relevance of the research.
- **Background summary/Introduction**– identify the study area your proposed thesis falls into, explaining its significance, likely impact and how it will contribute to the field.
- **Aims/Objectives**– explain the key aims of your research and the questions you will address in order to achieve them, state your hypothesis and/or research questions.
- **Methods**– detail how you intend to carry out your research (study population, design, exams/measurements, statistical analysis, sample size, field studies, interviews) and how you will analyse the data you collect.
- **Expected results & Data/Statistical analysis** (if required)
- **Time plan with milestones** - a schedule of key activities can help demonstrate that your research is achievable within the timeframe of your study.
- **Bibliography**– list the publications and sources you have cited in your proposal – you can also include sources you have read in preparation.

Very often, the aim of a proposal is to apply for **funding**. Refer to the infographic on the next page for some tips for writing such a proposal.

15 KEY TIPS FOR WRITING A WINNING GRANT PROPOSAL



BEFORE YOU START WRITING...



LEARN MORE ABOUT THE FUNDING AGENCY

Read their website to understand their goals and how your research will help in achieving these goals



UNDERSTAND YOUR TARGET AUDIENCE

Read up about the reviewing committee and decide which aspects of your subject matter need to be explained in detail



GO THROUGH THE FUNDER'S GUIDELINES FOR PROPOSALS

Make a list of everything you need to do: which questions to address, sections to include, supporting documents to be submitted, formatting specifications to be considered

WHILE WRITING YOUR GRANT PROPOSAL, MAKE SURE YOU TALK ABOUT:



The usefulness of your research for scientific development and human knowledge



How the sample used in the study is widely applicable



What motivated you to choose your research topic



The expertise and experience of all the researchers who are part of your study



How your work is aligned with the funding institution's mission and vision



Details of how your institution is equipped to support you, and the facilities it offers to help your research



The problems you have identified and the solutions your study will provide



The resources you need or plan to use to conduct your research



The credibility of the methods you plan to use



Details of how your study is in compliance with relevant ethical standards



How your study has a focused and measurable aim



How you are equipped to lead your research team well

3. The Introduction section of a research article or thesis

In the introduction section, one sets the stage for the rest of the article/thesis. Here one needs to catch the reader's attention and assert the fact that one's study is valuable by using strong terms such as 'central', 'crucial', 'problematic', 'particularly interesting'. This '**centrality claim**' shows that the general research area is important, central, interesting, problematic or relevant in some way. It is important to point out that one's research is linked to current thinking in the field of study. That is done by discussing recent literature in the **literature review** (see script Lesson 3). However, one should then point out that there is an **unresolved problem** or **gap** in the current research that one's study tends to fill, which is the justification for the research project. The final statement shows how the study will address the gap/problem.

One can identify three 'moves' with steps that are included in research introductions:

Move 1: Establishing a territory

This is generally accomplished in two ways: by demonstrating that a general area of research is important, critical, interesting, problematic, relevant, or otherwise worthy of investigation, and by introducing and reviewing key sources of prior research in that area to show where gaps exist or where prior research has been inadequate in addressing the research problem.

- Step 1 Claiming centrality and/or
- Step 2 Making topic generalizations and/or
- Step 3 Reviewing items of previous research

Move 2: Establishing a niche

Here you argue that your piece of research is important and possesses value. This can be done by indicating a specific gap in previous research, by challenging a broadly accepted assumption, by raising a question, a hypothesis, or need, or by extending previous knowledge in some way.

- Step 1 (a) Indicating a gap, or (b) making a counterclaim, or (c) raising a question, or (d) continuing a tradition

Move 3: Occupying the niche

The final "move" is to announce how your study will contribute new knowledge or new understanding in contrast to prior research on the topic. This is also where you can describe the remaining organizational structure of the paper.

- Step 1 (a) Outlining aim/purposes, or (b) Announcing present research
- Step 2 (possible) Announcing principal findings
- Step 3 Indicating Research article structure

Adapted from Swales' CARS: Creating a Research Space model of research introductions.
Source: <http://libguides.usc.edu/writingguide/CARS>

[Type here]

The following visual (Busse and August 2020) illustrates the structure of an Introduction:



3.1 Establishing a territory: Centrality statement

In the centrality statement, the first sentence or first few sentences) of an Introduction section, the author 'states that the topic of the research is useful, relevant, important, or worth investigating since it forms part of a lively, significant or well-established **research** area. **Centrality** claims frequently serve as topic sentences and **are** therefore usually followed by evidence to support this **statement**.' (Swales and Feak: CARS model of writing an introduction'

Examine the introduction section of an article or thesis.

- Does the writer **claim centrality/importance** for his/her research? If so, which words are used to refer to the importance of the research? Write them down. Then compare the words you have found with words used by other writers in your fellow students' texts.

When referring to the **centrality/importance of a research study**, writers often use certain phrases. Swales and Feak (2012:339) and the Academic Phrasebank list some of the phrases that are used frequently. They include phrases such as:

... has been extensively studied
... there has been growing interest in...
Recent studies have focused on...
... has become a major issue...
... remains a serious problem...
... there has been increasing concern...
... has been investigated by many researchers...
... has become an important aspect of...
X is fundamental to ...
X has a pivotal role in ...
X is frequently prescribed for ...
X is fast becoming a key instrument in ...
X plays a vital role in the metabolism of ...
X plays a critical role in the maintenance of ...
Xs have emerged as powerful platforms for ...
There is evidence that X plays a crucial role in regulating ...
X is a common condition which has considerable impact on ...
In the new global economy, X has become a central issue for ...
Evidence suggests that X is among the most important factors for ...
X is important for a wide range of scientific and industrial processes.
Xs are one of the most widely used groups of antibacterial agents and ...
There is a growing body of literature that recognises the importance of ...

3.2 Establishing a niche: Gap/Problem statement

In the introduction to your thesis, it is essential that you indicate that your research is valuable by pointing out a problem or gap in previous research or by extending previous research in some way. This statement provides the motivation for doing your study and criticizes the fact that something important has been omitted from research so far. Examples of such sentences could be:

- 'However, little is known about students' views of the use of tablet PCs in class.'
- Or
- 'Apart from a small study by Cooper (2012), this phenomenon has not been the focus of research so far.'

Consider some articles or theses.

- Can you find a sentence or paragraph that defines the gap in the literature that establishes the niche for the research undertaken in this study? Write it down.

[Type here]

- Compare this statement to those used in other students' texts. How is the problem/gap expressed in other texts?

When making a **gap or problem statement**, one writes what is still lacking in the field, and why it is important to address that problem. For example, "However, [topic] is still poorly understood (or under-examined or excluded or misinterpreted). This lack of attention is significant because [knowing about this topic] will provide a benefit OR [not knowing about this topic] will incur a cost".

Certain phrases are frequently used in problem/gap statements. Consider this example:

"... The relationship between the four factors, i.e., has not yet been clarified, so it is not clear which one is dominant in"

In order to indicate a gap in the research, authors often use what is termed '**quasi-negative statements**', using words such as 'little', 'few', 'inadequate', 'no/none', such as the following:

- However, little information.. (not enough)
- Little work.... Little data... Little research...
- Few studies/investigations... Few researchers... Few attempts...
- No studies/calculations/data to date have...
- None of these studies/calculations have...



Be sure of your facts; else add 'To the best of our/my knowledge, no studies...'

Language note: 'Little' is used with uncountable nouns, whereas 'few' is used with countable nouns. They both mean 'not enough'. ('a few' or 'a little' means 'some'.)

This table contains some useful verbs and adjectives that could help you formulate your gap statement. You could try to incorporate them into your gap statement (Swales & Feak, 2012:351).

[Type here]

VERBS	ADJECTIVES
Disregarded	Controversial
Ignored	Incomplete
Failed to consider	Inconclusive
Concentrated on	Questionable
Been limited to	Unconvincing
Misinterpreted	Unsatisfactory
Neglected to consider	Ambivalent
Overestimated	
Underestimated	
Overlooked	
Been restricted to	
Suffered from	

The Academic Phrasebank contains many phrases that might help you identify (and write) a problem or gap statement:

...has/have not been... (studied/reported/elucidated)

...is required/needed...

...the key question is/remains...

...it is important to address...

Previous studies of X have not dealt with ...

Researchers have not treated X in much detail.

Such explanations are unsatisfactory because they ...

Most studies in the field of X have only focused on ...

Such approaches, however, have failed to address ...

Previous published studies are limited to local surveys.

Half of the studies evaluated failed to specify whether ...

There is little published data on ...

No previous study has investigated X.

The use of X has not been investigated.

There has been no detailed investigation of ...

There has been little quantitative analysis of ...

Data about the efficacy and safety of X are limited.

Up to now, far too little attention has been paid to ...

A search of the literature revealed few studies which ...

The impact of X on Y is understudied, particularly for ...

Few studies have investigated X in any systematic way ...

In addition, no research has been found that surveyed ...

So far, however, there has been little discussion about ...

[Type here]

So far, very little attention has been paid to the role of X.

The research to date has tended to focus on X rather than Y.

Previously published studies on the effect of X are not consistent.

Statements indicating a knowledge gap in the field of study often begin with the word '**However**', for example:

However, very little is known about X in...

However, few studies have investigated ...

However, the nature of X remains unclear.

However, much less is known about how ...

However, the use of X has not been investigated.

However, far too little attention has been paid to ...

However, the behaviour of X has not yet been investigated.

However, the evidence for this relationship is inconclusive ...

However, much uncertainty still exists about the relation between ...

However, there have been no controlled studies which compare differences in...

Examine the Introductions of the articles you brought with you.

- Identify the centrality claim and gap/ problem statement.
- Were any of the above or similar phrases used? Write them down.

3.3 Occupying the niche: Filling the gap/addressing the problem

Once one has indicated that there is a gap in the research, one has to **show how one's study will fill that gap** – how it will occupy the niche. Here the authors indicate the main purpose or main feature of the research. Such purpose statements mostly refer to the **type of text** or genre, such as paper, article, dissertation, thesis, report, etc., or to the **type of investigation**, such as experiment, investigation, study, survey, etc., rather than to the writer (you) in the first person, although that differs from one field of study to the next. Increasingly, 'We' is used in these statements, for example 'Here we show a novel method of isolating...'

Examine some articles or theses.

- Can you find a sentence or paragraph that makes a statement about the purpose of the research: how the research aims to fill the gap in literature? Write it down.

- Compare this statement to those used in other students' texts. Write down different ways that are used to describe how the gap will be filled by your research.

- Is the first person singular (I) or plural (we) used or does the purpose statement refer to the type of text (meta-discourse)?

Some phrases that can be used to indicate this **gap "fill"** are:

- "We therefore analysed..."
- "In this study, we investigated..."
- "Therefore, the goals of this study are..."
- "In this paper, we report..."

Consider the following list of statements (adapted from Swales and Feak, 2012:355) that can be used to describe the purpose of the study and fill the research gap. Incorporate them into your writing if you find them appropriate.

- The aim of this research/investigation is to give...
- This study reports on the results obtained...
- The main purpose of the experiment reported here was to...
- This study was designed to evaluate...
- The present work extends the last model...
- The primary focus of this dissertation is..
- The primary objective of this paper is to provide...

[Type here]

Some more phrases used to indicate how a study fills a gap/addresses the problem in the research are (see Academic Phrasebank for more examples):

This work will generate fresh insight into ...

The study offers some important insights into ...

Understanding the link between X and Y will help ...

This is the first study to undertake a longitudinal analysis of ...

The present research explores, for the first time, the effects of ...

The importance and originality of this study are that it explores ...

The findings should make an important contribution to the field of

Characterisation of X is important for our increased understanding of ...

It is hoped that this research will contribute to a deeper understanding of ...

This study aims to contribute to this growing area of research by exploring ...

This project provided an important opportunity to advance the understanding of ...

Therefore, this study makes a major contribution to research on X by demonstrating ...

There are several important areas where this study makes an original contribution to ...

The experimental work presented here provides one of the first investigations into how ...

Now **write your own centrality claim, gap statement and statement to fill the gap** for your article or thesis. Try to express it in different ways. You could use some of the phrases listed above.

4. The hypothesis / research questions

Researchers explore the topic using a hypothesis or research question(s). Natural scientists usually already have a research hypothesis before they start writing the article or thesis. However, while a hypothesis is frequently used in the natural sciences, it is not limited to those disciplines. A hypothesis can be applied to and tested for subjects ranging from the humanities to economics. A hypothesis statement can be defined as...

a proposed answer to a question or problem that can be verified or rejected through testing. A hypothesis statement is typically an educated guess as to the relationship between factors, and serves as the basis for an experiment to test whether the relationship holds true. [from [BusinessDictionary.com](https://www.businessdictionary.com/definition/hypothesis.html)]

The hypothesis clearly states what the researcher intends to investigate. It should be specified before research is conducted and openly stated in reporting the results. A hypothesis must include a **prediction of the anticipated outcome** of the experiment, and it must be **testable**.

The basic formulation of a hypothesis could be: "If (I do this) (=independent variable) then (this will happen) (=dependent variable)." The structure of the sentence can obviously get quite complicated, depending on the complexity of the theory itself, but it can also be very straightforward and simple. The independent variable is the one you, the scientist control and the dependent variable is the one that you observe and/or measure the results. Look at these simple examples of 'If... then...' hypotheses:

- If you get at least 6 hours of sleep, you will do better on tests than if you get less sleep.
- If you stop smoking, your lung capacity will improve within 6 months.
- If skin cancer is **related** to ultraviolet light, **then** people with a high exposure to ultraviolet light will have a higher frequency of skin cancer.
- If leaf colour change is **related** to temperature, **then** exposing plants to low temperatures will result in changes in leaf colour.
- If the rate of photosynthesis is **related** to wave lengths of light, **then** exposing a plant to different colours of light will produce different amounts of oxygen.
- If the volume of a gas is related to temperature, then increasing the temperature will increase the volume.

Analyse some introduction sections and compare your findings with fellow students.

- Does the thesis or article contain a hypothesis? If so, how is the hypothesis formulated?
- Does the hypothesis include a prediction of the expected outcome, and is it testable?
- If there is no hypothesis, are there any **research questions**? If so, how are the research questions formulated?
- Did the writer perhaps use a hypothesis as well as research questions?

[Type here]

In the humanities and some other fields, including the Natural Sciences, **research questions** are sometimes used rather than hypotheses, in qualitative as well as quantitative research. When formulating research questions, the words ‘what’ or ‘how’ are mostly used. The questions specify whether the study aims to discover something, seeks to understand a phenomenon, explores or describes experiences. They must be open-ended, that is, not have ‘yes’ or ‘no’ as an answer. The questions describe rather than compare or relate variables (as is done in a hypothesis). The questions tend to be reviewed and reformulated during the course of the study.

The research questions could take the form of direct or indirect questions, for example:

- Direct: How does the use of insecticide-treated bed nets (ITNs) affect malaria illness, severe disease, and death due to malaria in endemic regions?
- Indirect: This study investigates whether insecticide-treated bed nets (ITNs) affect malaria illness, severe disease, and death due to malaria in endemic regions.

As in the hypothesis, research questions use **meta-discourse** to refer to the study/type of text rather than to the researcher in the first person (I).

- Is that the case in the studies that you are examining? Compare your studies with those of the other students.

Note that research questions should be **open-ended**, that is, they should not have ‘yes’ or ‘no’ as an answer but invite answers that can be discussed. They could start with ‘how’, ‘why’, ‘which’, etc.

Compare these research questions, with explanations of their suitability:

Research question	Explanation
✗ Does the US or the UK have a better healthcare system? ✓ How do the US and the UK compare in health outcomes among low-income people with chronic illnesses?	The first question is too broad and subjective: there’s no clear criteria for what counts as “better.” It is also not open ended. The second question is much more researchable. It uses clearly defined terms and narrows its focus to a specific population.
✗ Does gaming have an effect on the mind? ✓ What effect does 2 hours of daily gaming have on the attention span of 12-16 year-olds?	The first question is not specific enough. It is also not open ended. The second question is more researchable using qualitative and quantitative data collection.

Now write hypotheses or research questions relating to your field of study.

[Type here]

- Formulate the problem that you want to investigate in your study as a hypothesis and then as a research question.
- Which format does your department or your supervisor require you to use?
- Which of the formats do you prefer? Why? Discuss your answer with your fellow students.

Assignment 4 (Reviewed assignment)

- Write 3 different versions of possible introductory sentences for your thesis/ proposal/ article. (See the lesson slides for ideas.)
- Write a 250-350-word **introduction section** to the manuscript (e.g., proposal, article, thesis) you plan to work on for the remainder of the course. Include the following elements, **highlighted** in **different colours** in your text, and **identified** in the margin.
- A suitable introduction statement
- A very brief summary of or reference to some relevant literature
- A statement defining the problem or gap in the research/literature
- Your research hypothesis or research questions, or the study goal(s).
- A statement or paragraph in which you show how you intend to fill the gap/niche in research.
- Reference all sources used in the text and in a short bibliography.
- Review a colleague's introduction section.

Put your paragraph through the plagiarism check, Turnitin, on the ADAM page. Upload the Turnitin report too (not the receipt).

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[Type here]

University of Nottingham. N.d. How to write a research proposal.

<https://www.nottingham.ac.uk/pgstudy/how-to-apply/how-to-write-a-research-proposal.aspx>

Example Introduction:

If you do not have any articles with you, you could also consider the text below:

Vaccines have transformed medicine and dramatically improved human health. However, traditional vaccines that consist of attenuated or killed pathogens are not always practical or effective. A more modern strategy is to present specific immunogenic subunits from a pathogen to the immune system [1], [2]. Antibodies raised to these subunit vaccines, which are generally recombinant proteins or synthesized peptides, recognize native proteins from the appropriate pathogen and confer protection to the vaccinated individual. This reductionist approach is generally safer than traditional vaccines because there is no risk of a recombinant protein becoming virulent, and protein subunits promote a more directed immune response because of decreased antigenic competition. In the United States, vaccines cost \$10–120 per dose (Center for Disease Control; [3]) with subunit vaccines being the most expensive. Though impoverished nations would benefit the most from potential subunit vaccines for malaria, dengue, and other diseases for which traditional vaccines do not exist, the cost of recombinant subunit vaccines is too expensive for use in these areas [4]. Thus, cheaper methods for producing subunit vaccines should be a priority if we are to reduce the heavy disease burdens found in low-income countries. (<http://journals.plos.org/plosone/article?id=10.1371/journal.pone.0037179>)